Lab exercise 1: Plant Anatomy

Introduction

To learn about plant physiology, you need some familiarity with plant anatomy. If physiology is a collection of verbs (processes), then the parts of plants from the cell to the whole plant are the nouns that these verbs act on.

In this lab exercise you will use microscopes to study plant cells, tissues, and organs. Prepared slides using various stains will help you to see the cellular organization of roots, stems, and leaves, and some of the details of xylem and phloem cells. You may also prepare a few wet mounts of living material from plants you can collect from outside.

There will be no introductory "lecture" to this lab exercise – you can simply come in and get to work. Lab staff will be available to help. You and your partner will be called into the greenhouse for 5-10 minutes to do your part in setting up the nutrition experiment (Lab Exercise 3).

Learning objectives

- *Recognise different plant cells, tissues, and organ structures*
- *Associate plant anatomical structures to their physiological functions*
- *Obtain hands-on experience on working with microscopes and sample preparation*

Procedure

Download and/or print out a copy of the worksheets before coming to lab.

Prepared slides

- You and your partner have been given a box containing a selection of prepared microscope slides corresponding to most, but not all, of the photos on the worksheets (different groups may have different slides). The plant type and part are identified on the worksheets and indicated on the slide labels.
- Go through the slides in your box and identify the type of section (xs = cross section/tverrsnitt, ls = longitudinal section/lengdesnitt) and approximate magnification (40, 100, or 200x) of the corresponding photos on the worksheet. Write this information in the comments field on the worksheet.
- Find and label the structures listed in Table 1 below. Write the names of the structures using arrows, circles, ovals, or other shapes to indicate the location of the structure. Pencil is recommended. If there are photos of a structure at more than one level of magnification, you only need to label a structure once, at the most appropriate magnification.
- Use the images in the textbook to help you find the features – check the textbook before you ask for help! You can also try googling the name of the structure to find definitions and examples.
- Write brief answers to the questions on page 2 of the worksheet. Provide a clear, concise and well-formulated answer that addresses the question, and remember you have a maximum of 50 words per answer. Include sufficient detail to demonstrate understanding of the topic without including unnecessary or confusing information. Use appropriate, scientific language.
- **You only need to label the structures for the slides you find in your box**, but if you are curious about a photo from a slide that is not in your box, you are welcome to borrow slides from other groups, as long as you return them to the original box.

- We have included a few slides (*Elodea* meristem, *Zea* seed) that you do not need to label but illustrate other structures that you will encounter later in the course. If you have time, take a look at these and identify the structures that you see.

Wet mounts

- If you still have time, go outside and collect plant leaves you would like to see under the microscope (inform your lab leader before leaving so they know you are not finished yet). Label the structures that you can see on your worksheet.

Lab report **Deadline on September 3rd at 4pm on Canvas**

For this lab exercise, the lab report is the worksheet where you will name the structures you see with the microscope. **You will submit an electronic copy of the worksheet on Canvas by September 3rd at 4pm**. The worksheets will be evaluated for correct identification of the structures and sufficient answers to the questions. You and your lab partner must hand in one copy with both names on it. <u>We will not accept hand-ins of the printed worksheet.</u>

Assigned points:

- Each box should have 11 slides, so your lab reports should contain at least 11 labelled sections.
- Each slide counts for 5.45 points, so the 11 slides account for 60 points of the lab report.
- Each question counts for 10 points (30 points in total)
- 10 points will be given to overall appearance and readability of the report
- Structures should be well indicated, with clear arrows. They can be hand-written as long as they are clear

Notes

Some common stains

Phloroglucinol stains lignin in the secondary cell walls of xylem elements and other sclerenchyma cells **red**

Safranin red stains cutin, chromatin, lignin, phenolics, tannins, and chloroplasts **red**

Fast Green stains cellulose in primary cell walls **green**

Plant genera and English and Norwegian common names

Ammophila = dune grass, sandrør (xerophyte = drought tolerant)

Coleus = praktspragle

Cucurbita = cucumber, agurk or pumpkin, gresskar

Elodea = vannpest

Ficus = fig, fiken (succulent)

Helianthus = sunflower, solsikke

Nymphaea = water lily, vannlilje (floating aquatic)

Picea = spruce, gran

Pinguicula = butterwort, tettegrass (insectivorous)

Pinus = pine, furu

Rhoeo = oyster plant

Syringa = lilac, syrin

Tilia = linden, lind

Vicia = bean, bønn

Urtica = nettle, nesle

Zea = maize, mais (C4 grass)

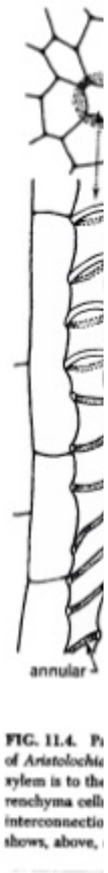
Table 1. Anatomical features visible in prepared microscope slides.

Slide	Root cap	Root meristem	Columella	Vascular tissue	Ground tissue	Endodermis	Casparian strip	Pericycle	Epidermis	Cortex	Phloem	Xylem	Schlerenchyma/phloem	Oxalate crystal	Collenchyma	Vascular bundle	Sieve cell	Companion cell	Sieve plate	Vessel element	Tracheids	Xylem fibers	Simple pits+	Boredered pits	Spiral thickening	Vascular ray	Pith	Vascular cambium*	Cork cambium*	Resin canal	Stomata	Spongy mesophyll	Palisade mesophyll	Chloroplasts	Aerenchyma	Sclerids	Hydrenchyma	Trichomes	C4 bundle sheath	Bulliform cells	Apical meristem	Leaf primordia	Seed coat	Endosperm	Embryo	Aleurone
Vicia faba (bønne) root tip ls	X	X	X	X	X				X																																					
Ranunculus (solleie) root xs						X	X	X	X	X	X	X																																		
Iris root xs						X	X	X	X	X	X	X		X																																
Zea mays (mais) stem xs & ls									X	X	X	X				X	X	X		X																										
Helianthus (solsikke) stem xs & ls									X	X	X	X	X			X	X	X	X	X			X																							
Cucurbita (agurk) stem xs & ls									X	X	X	X	X		X	X	X	X	X	X			X		X																					
Picea (gran) stem xs, ls, & ts											X	X									X		X	X		X	X	X	X	X																
Pinus (furu) stem xs, ls, & ts											X	X									X		X	X		X	X	X	X	X																
Tilia (lind) stem xs											X	X								X		X	X			X	X	X	X																	
Betula (bjørk) stem xs & ls											X	X								X		X	X		X	X	X	X	X																	
Pinus sylvestris (furu) leaf xs									X							X														X	X	X														
Syringa (syrin) leaf xs									X							X															X	X	X													
Nymphaea (vannlilje) leaf xs									X							X															X	X	X	X	X	X										
Ficus (fiken) leaf xs									X							X															X	X	X	X			X									
Ammophila (sandrør) leaf xs									X						X	X															X	X						X								
Zea (mais) leaf xs									X							X															X	X							X	X						
Elodea (vannpest) stem ls																																									X	X				
Zea (mais) seed ls																																											X	X	X	X

*The lateral cambia are single cell layers and hard to see directly, but you can indicate where they are based on their placement relative to other tissues.

+ simple pits are only visible in longitudinal sections

Figure 1. Examples of secondary wall types (“thickening” or “sculpturing”) in xylem elements.



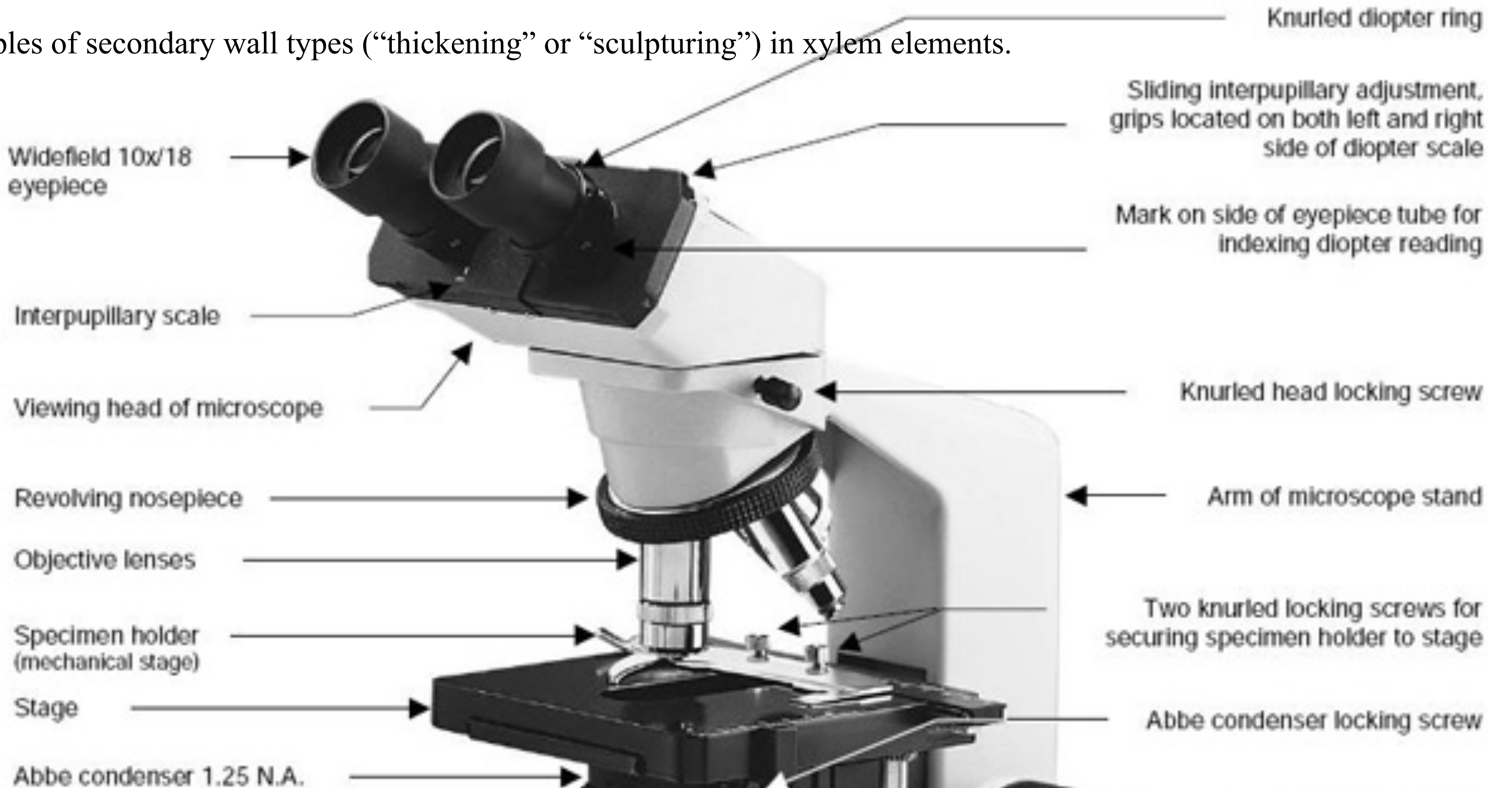


Figure 2. Cross-section of a young *Tilia* stem with detail showing location of wedges of primary and secondary tissues in the bark. (Campbell et al. 2011)